EXPERIMENT - 5

IPTables Firewall Configuration in Kali Linux

Step 1: Start Environment

Open Oracle VirtualBox and start Kali Linux virtual machine. Log in to the system.

Step 2: Open Terminal and Check IPTables

Open the terminal and verify that IPTables is installed and available in the system.

Step 3: View Existing Rules

Check the current firewall rules to understand the default configuration before making changes.

Step 4: Configure Firewall Rules

Set default policies to control incoming and outgoing traffic.
Allow necessary services such as localhost communication, SSH, HTTP, and HTTPS.
Block unwanted or suspicious IP addresses.

Step 5: Save and Verify Rules

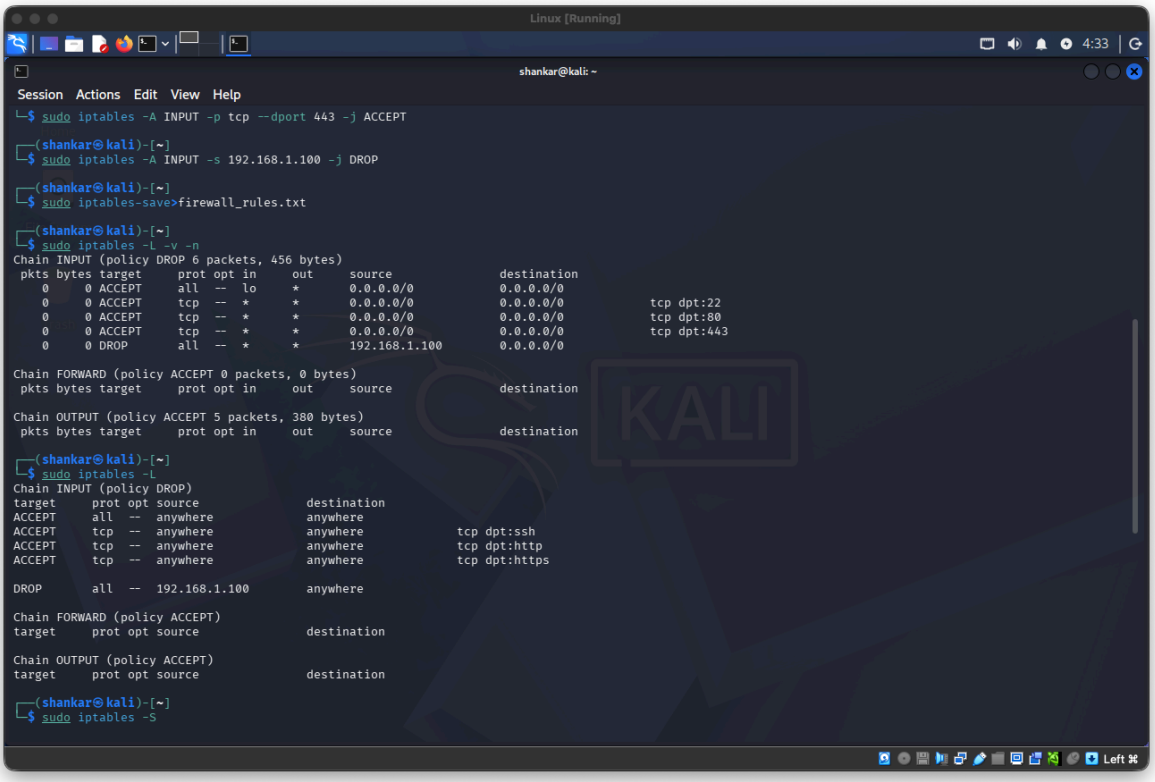
Save the configured firewall rules and verify that they are applied correctly by reviewing the rule list.

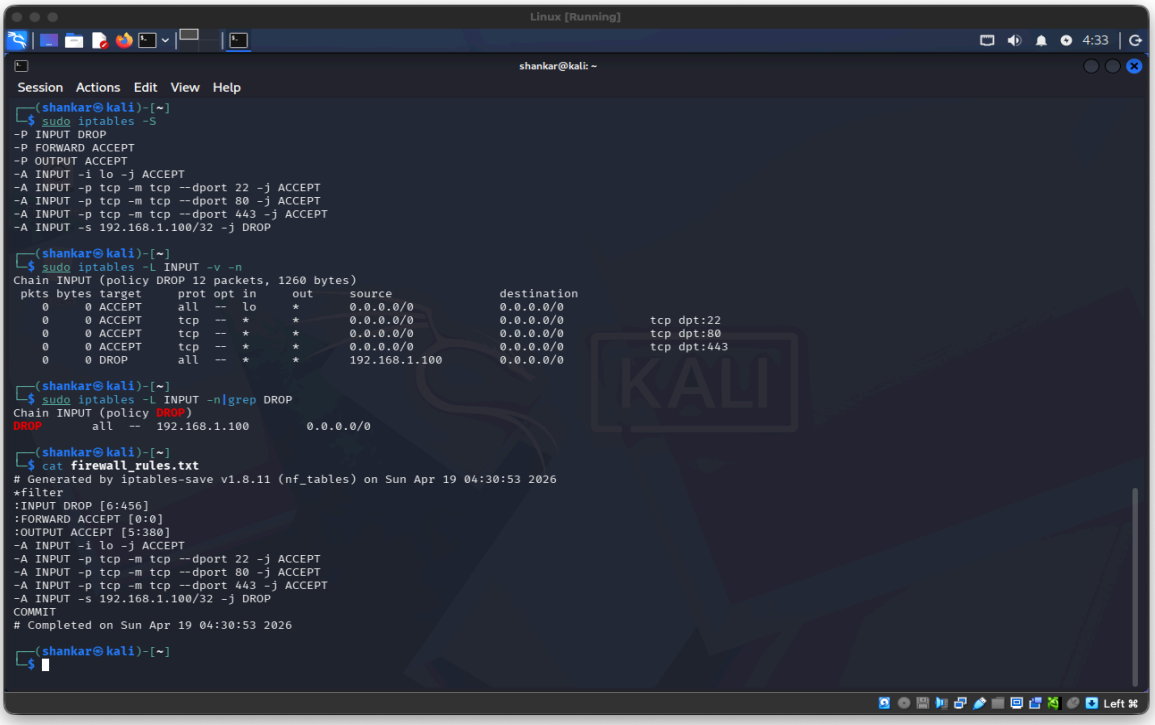
Linux [Running]

shankar@kali: ~

Session Actions Edit View Help

```
┌──(shankar㉿kali)-[~]
└─$ iptables
iptables v1.8.11 (nf_tables): no command specified
Try `iptables -h' or 'iptables --help' for more information.

┌──(shankar㉿kali)-[~]
└─$ iptables --version
iptables v1.8.11 (nf_tables)

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -:
[sudo] password for shankar:
sudo: a password is required

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -L
[sudo] password for shankar:
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -P INPUT DROP

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -P OUTPUT ACCEPT

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -i lo -j ACCEPT

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

┌──(shankar㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```





Result

Various firewall rules were implemented using IPTables, demonstrating how network traffic can be controlled to enhance system security.